

Constitution — Agent Failure Bench

Non-negotiable principles for this project. Every document and experiment must comply. Amendments to this document require an explicit, dated changelog entry at the bottom.

Purpose

This repository answers the research question: *How can a benchmark be designed to both measure the performance of agentic AI systems and systematically explain the causes of their failures?* The repository is **documentation only** — research questions, taxonomy, guidelines, and experiment designs. Every document must serve a research question; anything with no traceable link to one does not belong.

Principles

1. Design before execution

No experiment runs before its design is written down in `research/`: what is measured, on what data, with which metrics, and what decision the result feeds. Results that deviate from the design mean either the run or the design was wrong — record which, in the same change.

2. Taxonomy changes are versioned and empirically justified

The failure taxonomy (`research/taxonomy-v*.md`) is a research artifact, not a config file. Categories are added, removed, merged, or redefined **only** through a new version with a revision-log entry citing the empirical evidence (e.g., "category X unused across 42 trajectories", "escape hatch used 7 times with pattern Y").

3. Annotations are reproducible from raw data

Every annotation must reference a stored, immutable trajectory by id and event span. Raw harness output is preserved; any normalized form must be re-derivable from it. Never hand-edit trajectories.

4. Every experiment run is pinned

A run is only citable if its configuration is recorded: harness (Harbor) version, benchmark (Terminal-Bench) version and task list, agent name and version, LLM model id (including self-hosted weights + serving stack version), date, and any seeds or step/time budgets. Unpinned runs are exploratory and must not enter the dataset.

5. Failures are the object of study

Success rate is a summary statistic; the unit of analysis is the failed trajectory. Design decisions (sampling, model choice, tooling) optimize for making failures inspectable, classifiable, and comparable — not for maximizing benchmark scores.

6. Distinguish observation from interpretation

What happened (events, outputs, test results) is stored separately from judgments about it (annotations, labels, root-cause flags). An annotation always records its annotator (the judge model id). This separation is a precondition for the judge-agreement study (RQ2).

Changelog

- 2026-07-09: Initial version.
- 2026-07-12: Scope reset for the 3-week timeline. Repository reduced to documentation only (code, schemas, and specs removed; recoverable in git history). Manual annotation dropped — the sole annotator is an LLM judge validated against TRAIL's published expert annotations. Principle 1 reworded from "Spec before code" to "Design before execution"; principles 3 and 6 reworded to be implementation-free. Purpose section updated accordingly.
- 2026-07-12 (b): Human spot-check removed — the project contains **no** human annotation of any kind; judge validity rests solely on the TRAIL agreement study. Principle 6 amended (human annotator id dropped).
- 2026-07-12 (c): Repo narrowed to documents serving the research question currently in progress (RQ1). Experiment-design, TRAIL-mapping, and Harbor-format notes removed (recoverable in git history); they return when RQ2+ work starts. Principle 1's file reference generalized.